07/04/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

Redemption Test

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

(i) Duration of Test is 3 hrs.

(ii) The Test consists of 180 questions. The maximum marks are 720.

(iii) There are three parts in the question paper consisting of Physics, Chemistry, Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).

(iv) Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.

(v) Use blue/black ballpoint pen only to darken the appropriate circle.

(vi) Mark should be dark and completely fill the circle.

(vii) Dark only one circle for each entry.

(viii) Dark the circle in the space provided only.

(ix) Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A charge particle q_1 is at position (2, 2, 1). The electrostatic force on another charged particle q_2 at (0, 0, 0) due to q_1, is

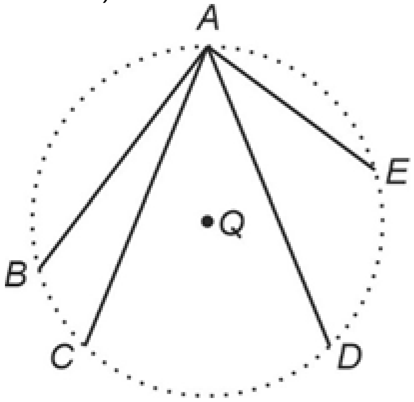
(1) $\frac{q_1 q_2}{108\pi\varepsilon_0}(2\hat{i}+2\hat{j}+\hat{k})$

(2) $\frac{q_1 q_2}{64\pi\varepsilon_0}(2\hat{i}+2\hat{j}+\hat{k})$

(3) $\frac{-q_1 q_2}{108\pi\varepsilon_0}(2\hat{i}+2\hat{j}+\hat{k})$

(4) $\frac{-q_1 q_2}{54\pi\varepsilon_0}(2\hat{i}+2\hat{j}+\hat{k})$

2. In the electric field of a point charge *Q*, another charge is carried from *A* to *B*, *A* to *C*, *A* to *D* and *A* to *E* as shown in figure, then work done will be (*Q* is placed at the centre of the circle)



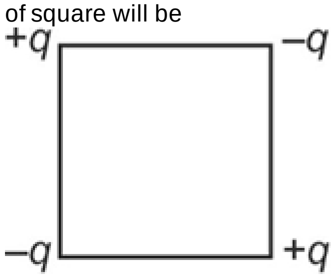
(1) Maximum along path *AB*

(2) Maximum along path *AD*

(3) Maximum along path *AE*

(4) Zero along all the paths

3. Four charges are arranged at the corners of a square of side $\sqrt{2}a$ as shown. The net electric potential at the centre of square will be



(1) $\frac{q}{\pi\varepsilon_0 a}$

(2) $\frac{q}{4\pi\varepsilon_0 a}$

(3) $\frac{q}{2\sqrt{2}\pi\varepsilon_0 a}$

(4) Zero

4. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved

(1) Remains the same

(2) Becomes eight times

(3) Becomes four times

(4) Becomes two times

5. A rod of length 50 cm is moving at a speed of 10 m/s in a magnetic field of strength 0.4 T. The emf induced in the rod will be

(1) 2 V

(2) 20 V

(3) 0.2 V

(4) Insufficient information

6. Which of the following has negative value of magnetic permeability?

(1) Diamagnetic substances

(2) Paramagnetic substances

(3) Ferromagnetic substances

(4) Magnetic permeability can't be negative

7. Resistivity of copper

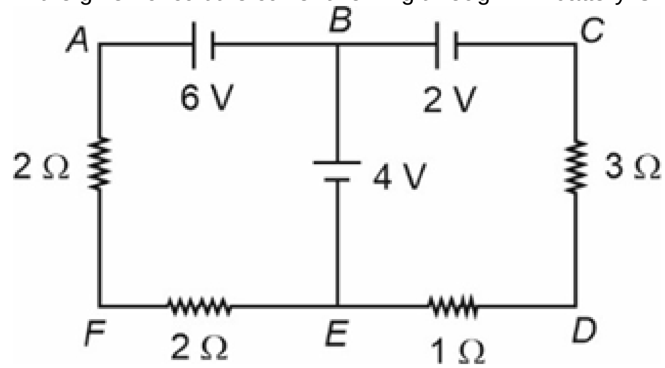
(1) Increases linearly with increase in temperature

(2) Increases non-linearly with increase in temperature

(3) Remains constant with increase in temperature

(4) Decreases non-linearly with increase in temperature

8. In the given circuit the current flowing through 4 V battery is



(1) 1 A

(2) 2 A

(3) 3 A

(4) 4 A

9. A long solenoid having turn density of 1000 turns per meter has a resistance of 100 Ω in its coil. If its ends are connected to a source of 1 kV then maximum value of magnetic field at its centre will be

(1) 4π mT

(2) 2π mT

(3) 8π mT

(4) $4\pi \times 10^{-3}$mT

10. Given below are two statements:
Statement I : The 220 V of domestic AC circuit represents peak voltage.
Statement II : Average value of sinusoidal AC current may have zero value in half time period. (where symbols have their usual meaning).
In the light of the above statements, choose the most appropriate answer from the options given below.

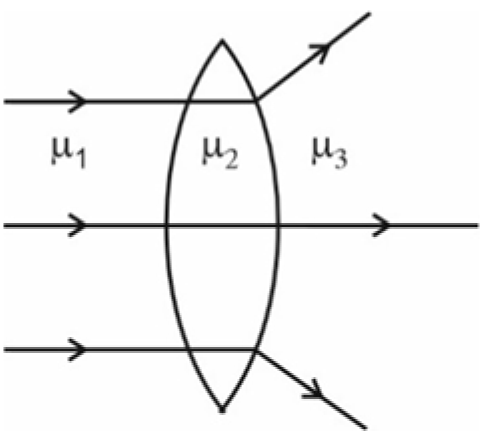
(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

11. Behaviour of light rays going through a lens is shown below.



The correct relation between μ_1, μ_2 and μ_3 is

(1) $\mu_1 > \mu_2 > \mu_3$

(2) $\mu_3 > \mu_2 > \mu_1$

(3) $\mu_2 > \mu_3$ and $\mu_1 = \mu_2$

(4) $\mu_2 < \mu_3$ and $\mu_1 = \mu_2$

12. After refraction through concave lens, plane wavefront becomes

(1) Remains planar

(2) Spherical

(3) Cylindrical

(4) Conical

13. For a series *LCR* circuit,
(a) Voltage across *L* and *R* differ by $\frac{\pi}{2}$ phase
(b) Current through *C* and *R* are in same phase
(c) Voltage across *C* and current through *L* differ by $\frac{\pi}{2}$ phase
The correct option with regard to above statements is

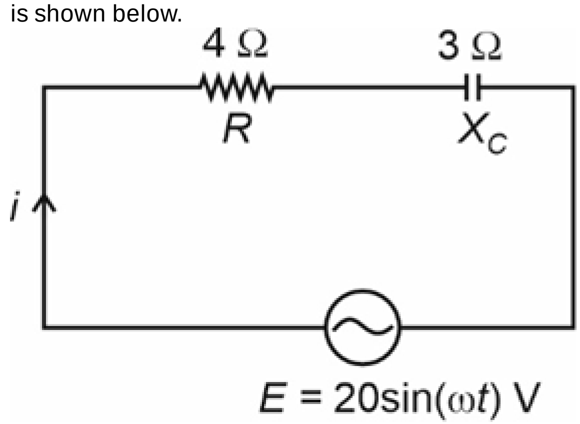
(1) Only (a) is true

(2) Only (a) and (b) are true

(3) Only (b) and (c) are true

(4) (a), (b) and (c) are true

14. An AC-generator supplying voltage *E* to a circuit that has a resistor of resistance 4 Ω and a capacitor of reactance 3 Ω is shown below.



Then voltage across capacitor is given by

(1) 12sin(ωt – 37º) V

(2) 16sin(ωt – 37º) V

(3) 12sin(ωt – 53º) V

(4) 16sin(ωt – 53º) V

15. A plane electromagnetic wave of energy *E* is incident normally on a plane surface of area *A* in time *t* such that it is 100% absorbed. If *c* is the speed of light in free space, then average pressure exerted on the surface is

(1) $\frac{2E}{c}$

(2) $\frac{2Et}{Ac}$

(3) $\frac{2E}{ctA}$

(4) $\frac{E}{ctA}$

16. The shape of meniscus when contact angle is less than 90° is

(1) Concave downwards

(2) Concave upwards

(3) Can either be concave downwards or concave upwards

(4) Insufficient information

17. In a photoelectric experiment, the frequency and photon intensity of a light source are both quadrupled. Then consider the following statements
(i) The saturation photocurrent remains almost the same.
(ii) The maximum kinetic energy of the photoelectrons is quadrupled.
The correct option is

(1) (i) is true but (ii) is false

(2) (i) is false but (ii) is true

(3) Both (i) and (ii) are true

(4) Both (i) and (ii) are false

18. The mass of a $^{11}_{5}B$ nucleus is 0.077 u less than the sum of the masses of all its nucleons. The binding energy per nucleon of $^{11}_{5}B$ nucleus is
(Given, 1 u × c^2 = 930 MeV)

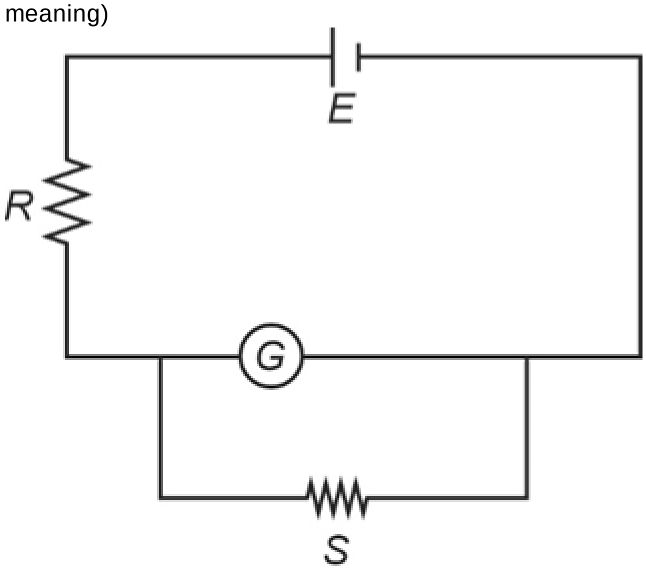
(1) 71.61 MeV

(2) 65.11 MeV

(3) 7.16 MeV

(4) 6.51 MeV

19. In an experiment to determine the resistance of galvanometer by half-deflection method, we use the following circuit where R = 600 Ω and G = 3 Ω. Then the value of resistance S is (where symbols have their usual meaning)



(1) $\frac{600}{201}\ \Omega$

(2) 3 Ω

(3) $\frac{600}{199}\ \Omega$

(4) 2 Ω

20. Consider the following statements regarding nuclear force and select the correct statement(s).
(I) They are independent of charge
(II) They are always attractive in nature
(III) They are non-central forces

(1) I only

(2) I and II

(3) I and III

(4) I, II and III

21. An external voltage (V) is applied across a diode such that it is reverse biased. As the reverse bias voltage is gradually increased before it reaches breakdown voltage, then current

(1) Sharply increases

(2) Sharply decreases

(3) First increases and then decreases

(4) Remains voltage independent

22. If kinetic energy of an electron in first orbit of hydrogen atom is K, then potential energy of an electron in first orbit of singly ionized helium atom is

(1) – 2K

(2) – 4K

(3) –8K

(4) – 16K

23. The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is perpendicular to the vector $\vec{P}$ and its magnitude is equal to half of the magnitude of vector $\vec{Q}$. Angle between $\vec{P}$ and $\vec{Q}$ is

(1) θ = 120°

(2) θ = 90°

(3) θ = 180°

(4) θ = 150°

24. A man swims across a river with speed of 3 km/h (in still water), while a boat goes upstream with speed 4 km/h (in still water). The speed and direction in which the man appear to go to the boatman, if the speed of flowing water is 1 km/h, is

(1) 13 km/h, $\tan^{-1}\left(\frac{1}{12}\right)$ with the direction of river flow

(2) 12 km/h, $\tan^{-1}\left(\frac{1}{3}\right)$ with the direction of river flow

(3) 5 km/h, $\tan^{-1}\left(\frac{3}{5}\right)$ with the direction of river flow

(4) 5 km/h, 37° with the direction of river flow

25. The unit $\frac{\text{joule ohm}}{sec\text{volt}}$ is equivalent to

(1) ampere

(2) weber

(3) volt

(4) tesla

26. **Statement A :** Dimensions of a physical quantity are the powers (or exponents) to which the base quantities are raised to express that quantity.
Statement B : Only those physical quantities can be added or subtracted which have same dimensional formula.
Consider the given statements and choose the correct option.

(1) Statement A is true but B is false

(2) Statement A is false but B is true

(3) Both statements A and B are true

(4) Both statements A and B are false

27. A particle moves along a straight line such that its displacement from the origin varies with time *t* as $x = \sin t - \sqrt{3}\ \cos t$, here *x* is in meters and *t* is in seconds, then the instant when it crosses the origin for the first time, is

(1) $\frac{\pi}{3}$ s

(2) $\frac{\pi}{6}$ s

(3) $\frac{2\pi}{3}$ s

(4) $\frac{4\pi}{3}$ s

28. A thin circular ring of mass *M* and radius *R* is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity 'ω'. If a disc of same radius but half mass is placed gently on the ring co-axially, then the new angular velocity of the system is

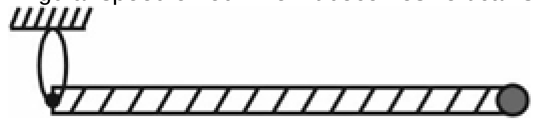
(1) $\frac{5}{4}\omega$

(2) $\frac{2}{3}\omega$

(3) $\frac{4}{5}\omega$

(4) ω

29. A uniform rod of mass 2*m* and length *L* with a small ball of mass *m* attached at one end is released from horizontal position as shown in figure. Other end of the rod is hinged. Angular speed of rod when it becomes vertical is



(1) $\sqrt{\frac{5g}{3L}}$

(2) $\sqrt{\frac{3g}{L}}$

(3) $\sqrt{\frac{12g}{5L}}$

(4) $\sqrt{\frac{3g}{2L}}$

30. A mass *m* is attached to a thin wire and whirled in a vertical circle. The wire is least likely to break when

(1) The mass is at the highest point

(2) The mass is at the lowest point

(3) The wire is horizontal

(4) Inclined at an angle of 45° with the vertical

31. Energy that will be ideally radiated by a 20 kW transmitter in 3 hour is

(1) 2.16×10^{8} eV

(2) 1.35×10^{27} J

(3) 1.35×10^{27} eV

(4) 21.6×10^{8} J

32. Ratio of the radius of gyration of a solid sphere (*M, R*) and a hollow sphere (*M, R*) about their tangential axis is [All the symbols have their usual meaning]

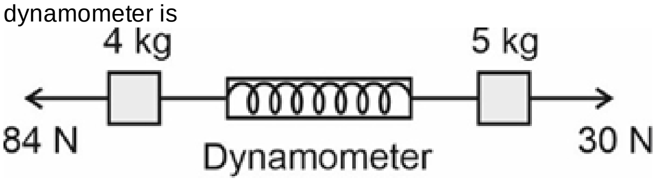
(1) $\sqrt{\frac{35}{15}}$

(2) $\sqrt{\frac{3}{5}}$

(3) $\sqrt{\frac{9}{4}}$

(4) $\sqrt{\frac{21}{25}}$

33. A dynamometer is attached to two blocks of masses 4 kg and 5 kg as shown in the figure. The reading of the dynamometer is



(1) 60 N

(2) 54 N

(3) 30 N

(4) 84 N

34. Consider two statements given below out of which one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): A thin stainless steel needle can lay floating on a still water surface.

Reason (R): Thin stainless steel needle floats when buoyancy force balances its weight.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are true and (R) is correct explanation of (A)

(4) Both (A) and (R) are false

35. A square shaped wire frame of side length R is floating on the surface of liquid of surface tension S. The force required to just pull out the frame from the liquid is

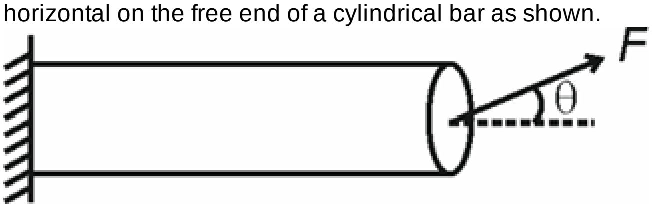
(1) $2SR$

(2) $4SR$

(3) $8SR$

(4) SR

36. A force of magnitude F is applied at an angle θ from horizontal on the free end of a cylindrical bar as shown.



The shear stress developed at the given cross-section of area A will be

(1) $\frac{F}{A}$

(2) $\frac{F\cos\theta}{A}$

(3) $\frac{F}{A\cos\theta}$

(4) $\frac{F\sin\theta}{A}$

37. Consider the following statements:

Statement (A): When height of capillary tube is insufficient the liquid does not overflow.

Statement (B): Product of radius of meniscus and height of liquid in capillary tube always remains constant.

In the light of above statements, choose the **correct** answer.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Statement (A) is correct while statement (B) is incorrect

(4) Statement (B) is correct while statement (A) is incorrect

38. The escape velocity of a 2 kg body on a planet is 100 m/s. The potential energy of body on that planet is

(1) – 10 kJ

(2) 5 kJ

(3) – 20 kJ

(4) +20 kJ

39. An ideal gas is compressed to half of its initial volume through different processes starting from same initial state. Which of the following process results in the maximum work done on the gas?

(1) Adiabatic

(2) Isothermal

(3) Isobaric

(4) Isochoric

40. Two simple harmonic motions are represented by the equations, $y_1 = 4\sin(\omega t)$ and $y_2 = 3\cos\left(\omega t + \frac{\pi}{2}\right)$. The amplitude of resultant SHM due to the superposition of these simple harmonic motions will be (All quantities are in SI unit)

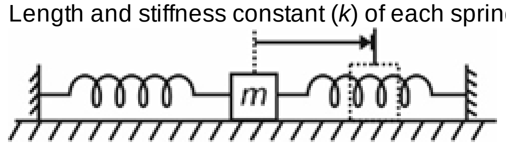
(1) Zero

(2) 1 m

(3) 5 m

(4) 7 m

41. A block is kept on a smooth surface in given arrangement. Length and stiffness constant (k) of each spring is same.



The block (of mass m) is pulled to the right by 5 m and then released from rest. Maximum speed reached by block will be (assume k = 10 N/m and m = 5 kg)

(1) 5 m/s

(2) $5\sqrt{2}$ m/s

(3) 10 m/s

(4) $10\sqrt{2}$ m/s

42. For a thermodynamic system, a diathermic wall allows flow of

(1) Heat only

(2) Matter only

(3) Heat as well as matter

(4) Neither heat nor matter

43. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: A combination of two simple harmonic motion with arbitrary amplitudes and phases is necessarily simple harmonic motion.
Reason R: A combination of two simple harmonic motions is periodic only if frequency of one motion is an integral multiple of other's frequency.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**

(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**

(3) **A** is true but **R** is false

(4) **A** is false but **R** is true

44. Longitudinal wave motion can be observed in

(1) Waves produced in a cylinder containing a air by moving its piston back and forth

(2) Waves produced by a motorboat sailing in water

(3) Motion of a kink in a longitudinal spring produced by displacing one end of the spring sideways

(4) All of these

45. Given below are two statements:
Statement I: At room temperature, all the gases behave as ideal gas.
Statement II: Real gas curve approach the ideal gas behaviour for high pressure and low temperature.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

CHEMISTRY

46. Volume of O_2 required at STP to react completely with 180 g of ethane is

(1) 470.4 L

(2) 448 L

(3) 134.4 L

(4) 224 L

47. Match the list I with list II.

List I (Amount)		**List II (Number of atoms)**
a. 7 g of nitrogen gas	(i)	0.1 N_A
b. 34 g of ammonia gas	(ii)	0.5 N_A
c. 1.6 g of oxygen gas	(iii)	2 N_A
d. 2 g of hydrogen gas	(iv)	8 N_A

Choose the correct match.

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(ii), b(iv), c(i), d(iii)

48. Consider the following two statements
Statement I : Bohr's theory was able to explain the splitting of spectral lines in the presence of magnetic filed or an electric field.
Statement II : Bohr's theory could not explain the ability of atoms to form molecules by chemical bonds.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Statement I is correct and statement II is incorrect

(3) Statement I is incorrect and statement II is correct

(4) Both statement I and statement II are incorrect

49. A 100 watt bulb emits monochromatic light of wavelength 6626 pm. The number of photons emitted per second by the bulb is (h = 6.626×10^{-34} Js)

(1) 3.33×10^{17}

(2) 3.33×10^{18}

(3) 3.33×10^{19}

(4) 3.33×10^{20}

50. An atom with which of the following electronic configuration has the highest ionization enthalpy?

(1) $1s^2 2s^2 2p^1$

(2) $1s^2 2s^2$

(3) $1s^2 2s^2 2p^3$

(4) $1s^2 2s^2 2p^4$

51. Consider the following statements
(a) Bi and Rb are representative elements
(b) Ge and Sb are semi-metals
(c) Ti and Ce are d-block elements
The correct statements are

(1) (a) and (c) only

(2) (b) and (c) only

(3) (a) and (b) only

(4) (a), (b) and (c)

52. Consider the following two statements
Statement I : Formal charges help in the selection of the lowest energy structure from a number of possible Lewis structures for a given species.
Statement II : Generally the, lowest energy structure is the one with the maximum formal charges on atoms.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

53. Following are the two statements, one is labelled as Assertion (A) and other is labelled as Reason (R)
Assertion (A) : The resultant dipole moment of NF_3 is greater than that of NH_3.
Reason (R) : Fluorine being more electronegative than hydrogen leads to greater dipole moment of NF_3 as compared to NH_3.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

54. Under isothermal conditions, the correct options for an ideal gas undergoing reversible compression is:

(1) $\Delta U = 0,\ \Delta S_{total} \neq 0$

(2) $\Delta U \neq 0,\ \Delta S_{total} = 0$

(3) $\Delta U \neq 0,\ \Delta S_{total} \neq 0$

(4) $\Delta U = 0,\ \Delta S_{total} = 0$

55. Given below are two statements:
Statement I: Heat added to a system at lower temperature causes greater randomness than when the same amount of heat is added to it at higher temperature.
Statement II: The total entropy change for system and surrounding of a spontaneous process is greater than zero.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

56. Given below are two statements: One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A): Aqueous solution of NH_4Cl will be acidic.
Reason (R): Aqueous NH_4Cl undergoes cationic hydrolysis.
In the light of above statements, choose the **correct** answer among the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

57. In which of the following solutions, $PbCl_2$ has minimum solubility?

(1) 0.02 M NaCl

(2) 0.02 M $MgCl_2$

(3) 0.01 M $PbSO_4$

(4) 0.02 M NH_4Cl

58. The oxidation state of Cr in CrO_5 is

(1) + 10

(2) + 8

(3) + 5

(4) + 6

59. The oxidation number of terminal Br in tribromooctaoxide is

(1) + 4

(2) + 7

(3) + 5

(4) + 6

60. Number of 3-centre-2-electron bond(s) in B_2H_6 is

(1) 3

(2) 2

(3) 4

(4) 1

61. Which among the following has highest melting point?

(1) B

(2) Al

(3) Ga

(4) In

62. Given below are the two statements
Statement I: Fractions of crude oil in petroleum industry are separated by fractional distillation.
Statement II: In fractional distillation, the vapours rising up in the fractionating column become richer in less volatile component.
In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

63. Given below are the two statements
Statement-I: Paper chromatography is a type of adsorption chromatography.
Statement-II: Adsorption chromatography is based on the fact that different compounds are adsorbed on an adsorbent to different degrees.
In light of above statements, choose the correct answer

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

64. Consider the following compounds
(i) Hexane
(ii) Heptane
(iii) 2, 2–Dimethylpropane
(iv) 2– Methylbutane
The correct order of their melting point is

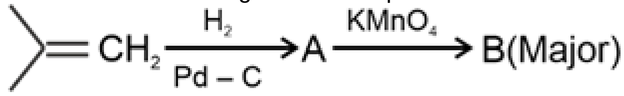
(1) (iii) > (ii) > (i) > (iv)

(2) (ii) > (i) > (iii) > (iv)

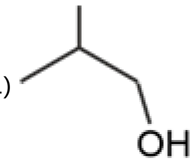
(3) (ii) > (i) > (iv) > (iii)

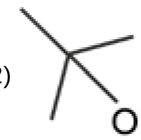
(4) (iii) > (iv) > (i) > (ii)

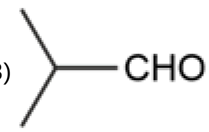
65. Consider the following reaction sequence

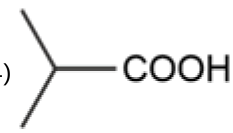


Major product B is

(1) 

(2) 

(3) 

(4) 

66. Given below are the two statements:
Statement I: The solutions which obey Raoult's law over the entire range of concentration are knwon as ideal solution.
Statement II: For ideal solution, $\Delta_{mix}H = 0$ and $\Delta_{mix}V = 0$.
In the light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

67. If mass percent (w/w%) of methanol in aqueous solution is 24% then the molality of methanol in the solution will be

(1) 4.25 m

(2) 9.87 m

(3) 6.71 m

(4) 7.85 m

68. Consider the change in oxidation state of iodine corresponding to different emf values as shown in the diagram below:

$H_3IO_6^{2-} \xrightarrow{0.7\,V} IO_3^- \xrightarrow{0.14\,V} IO^- \xrightarrow{0.45\,V} \frac{1}{2}I_2 \xrightarrow{0.54\,V} I^-$

The species which will undergo disproportionation are

(1) IO_3^- and I^-

(2) IO_3^- and IO^-

(3) IO^- and I_2

(4) IO_3^- and I_2

69. Given below are the two statements:
Statement I: Conductivity of iron is more than copper at 298.15 K.
Statement II: CuO acts as a semiconductor.
In the light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

70. Match list-I with list-II.

	List-I (Species)		List-II (Shape)
a.	SF_4	(i)	Tetrahedral
b.	NH_4^+	(ii)	Square planar
c.	BrF_3	(iii)	See-saw
d.	XeF_4	(iv)	Bent T-shape

Choose the correct answer from the options given below.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

71. Consider the following statements
Statement (I): For the reaction, A + B → Product, rate = $PZ_{AB}\, e^{-E_a/RT}$
Z_{AB} represents the collision frequency of reactants, A and B.
Statement (II): $e^{-E_a/RT}$ represents the fraction of molecules with energies equal to or less than activation energy (E_a).
In the light of above statements, choose the correct option.

(1) Both statement (I) and statement (II) are correct

(2) Statement (I) is correct but statement (II) is incorrect

(3) Statement (I) is incorrect but statement (II) is correct

(4) Both statement (I) and statement (II) are incorrect

72. The slope of ln k v/s $\frac{1}{T}$ of first order reaction is – 4 × 10^2 K. The value of E_a of the reaction is

(1) 33.25 kJ mol^{-1}

(2) 332.5 kJ mol^{-1}

(3) 3.325 kJ mol^{-1}

(4) 3325 kJ mol^{-1}

73. Match List-I with List-II

	List – I (Oxides of nitrogen)		List-II (Common methods of Preparation)
(a)	N_2O_5	(i)	$Pb(NO_3)_2 \xrightarrow{673K}$
(b)	NO_2	(ii)	$NH_4NO_3 \xrightarrow{Heat}$
(c)	N_2O	(iii)	$NaNO_2 + FeSO_4 + H_2SO_4 \rightarrow$
(d)	NO	(iv)	$HNO_3 + P_4O_{10} \rightarrow$

The correct match is

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

74. Which among the following products is formed when SO_2(g) reacts with Cl_2 in presence of charcoal?

(1) $SOCl_2$

(2) SO_3

(3) SO_2Cl_2

(4) S_2Cl_2

75. The colour and geometry of manganate ion is

(1) Green and tetrahedral

(2) Orange and pyramidal

(3) Purple and tetrahedral

(4) Pink and square planar

76. $KMnO_4$ on heating at 513 K gives

(1) K_2MnO_4 and O_2 only

(2) MnO_2, K_2MnO_4 and O_2

(3) MnO_2 and O_2 only

(4) K_2MnO_4 and MnO_2 only

77. Complex with maximum value of CFSE among the following is

(1) $[Fe(H_2O)_6]^{3+}$

(2) $[Fe(CN)_6]^{3-}$

(3) $[FeF_6]^{3-}$

(4) $[Fe(CN)_6]^{4-}$

78. Given below are two statements: one is labelled as **assertion (A)** and other is labelled as **reason (R).**
Assertion (A) : $H_2N - NH_2$ is a chelate ligand.
Reason (R) : In $H_2N - NH_2$ both nitrogen has one lone pair each.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

79. Correct order of leaving ability of groups is

(1) $\overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > F^{\ominus}$

(2) $F^{\ominus} > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3$

(3) $\overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3 > F^{\ominus} > \overset{\ominus}{O}H$

(4) $F^{\ominus} > \overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H$

80. $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_3 + \underset{(1\ eq)}{Cl_2} \xrightarrow{h\nu}$

Major product for above reaction will be

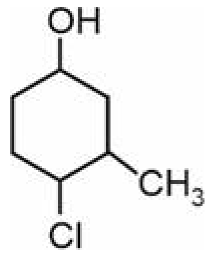
(1) $CH_3 - \overset{\overset{Cl}{|}}{\underset{\underset{CH_3}{|}}{C}} - CH_3$

(2) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_2 - Cl$

(3) $CH_3 - CH_2 - CH_2 - CH_2 - Cl$

(4) $CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_2 - CH_3$

81.



IUPAC name of compound given above is

(1) 1-Chloro-2-methylcyclohexan-4-ol

(2) 2-Chloro-1-methylcyclohexan-5-ol

(3) 4-Chloro-3-methylcyclohexan-1-ol

(4) 4-Chloro-5-methylcyclohexan-1-ol

82. $CH_3 - CH_2 - \overset{\overset{CH_3}{|}}{CH} - CH_2 - O - CH_2 - CH_3 + \underset{(1\ eq)}{HI} \longrightarrow$

Major products formed in reaction given above are

(1) $CH_3 - CH_2 - \overset{\overset{CH_3}{|}}{\underset{\underset{I}{|}}{C}} - CH_3 + C_2H_5OH$

(2) $CH_3 - CH_2 - \overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}} - CH_3 + C_2H_5 - I$

(3) $CH_3 - CH_2 - \overset{\overset{CH_3}{|}}{CH} - CH_2OH + C_2H_5 - I$

(4) $CH_3 - CH_2 - \overset{\overset{CH_3}{|}}{CH} - CH_2 - I + C_2H_5 - OH$

83. Consider the following carboxylic acids
(i) Lactic acid
(ii) Tartaric acid
(iii) Adipic acid
Which of the following can be categorised as optically active dicarboxylic acid?

(1) (i) and (iii) only

(2) (ii) only

(3) (ii) and (iii) only

(4) (iii) only

84. Consider the following compounds:
(i) Vanillin
(ii) Acrolein
(iii) Mesityl oxide
The compound(s) in which both –OH and –CHO groups are present is/are

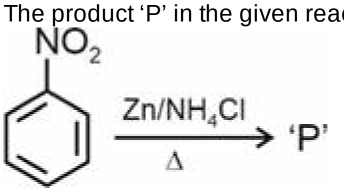
(1) (i) and (ii) only

(2) (i) only

(3) (ii) only

(4) (i), (ii) and (iii)

85. The product 'P' in the given reaction is



(1) N-Phenylhydroxylamine

(2) Azoxybenzene

(3) Azobenzene

(4) Hydrazobenzene

86. Following are the two statements labelled as Assertion (A) and Reason (R).
Assertion (A): Acetylation of $-NH_2$ group of aniline reduces its activating effect towards electrophilic substitution.
Reason (R): Acetyl group on benzene has electron withdrawing nature.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true and (R) is false

(4) (A) is false and (R) is true

87. Consider the following statements:
Statement-I: Pb^{2+} ions react with potassium chromate to form yellow precipitate.
Statement-II: Lead chromate is insoluble in hot sodium hydroxide solution.
In the light of above statements, choose the correct option.

(1) Statement-I is correct and Statement-II is incorrect

(2) Statement-I is incorrect but Statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both the statement-I and statement-II are incorrect

88. For the detection of functional group, alcohol (R – OH) is treated with ceric ammonium nitrate while phenol is treated with freshly prepared ferric chloride to obtain red and violet coloured complexes respectively. Chemically these red and violet complexes respectively are

(1) $[Ce(NO_3)_4(ROH)_3]$ and $[Fe(CN)_5NO]^{2-}$

(2) $[Ce(ROH)_4(NO_3)_3]$ and $[Fe(C_6H_5O)_6]^{3-}$

(3) $[Ce(ROH)_6(NO_3)_3]$ and $[Fe(CN)_5NO]^{2-}$

(4) $[Ce(NO_3)_4(ROH)_3]$ and $[Fe(C_6H_5O)_6]^{3-}$

89. The total number of chiral carbon atoms in α-D-(+)-Glucopyranose structure is

(1) 2

(2) 3

(3) 4

(4) 5

90. Consider the following statements:
Statement-I: Nucleotide contains a sugar molecule and a nitrogenous base only.
Statement-II: Nucleoside contains a sugar molecule, a nitrogenous base and a phosphoric acid molecule.
In the light of above statements, choose the correct option.

(1) Statement-I is correct but Statement-II is incorrect

(2) Statement-I is incorrect but Statement-II is correct

(3) Both the statements-I and II are correct

(4) Both the statements-I and II are incorrect

BIOLOGY

91. Outer layer of seed coat is

(1) Tegmen

(2) Testa

(3) Coleorhiza

(4) Funicle

92. Read the following statements of **Assertion** (A) and **Reason** (R) and choose the **correct** option.
Assertion (A): Ovules generally differentiate a single megaspore mother cell in nucellus.
Reason (R): Mostly, megaspore mother cell in a typical flowering plant forms only one megaspore.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

93. In a typical angiospermic embryo sac, the egg apparatus is

(1) 7-celled and 8-nucleate

(2) 8-celled and 7-nucleate

(3) 3-celled and 3-nucleate

(4) 7-celled and 3-nucleate

94. Chemoautotrophic bacteria of marshy habitats is a/an

(1) Facultative anaerobe

(2) Obligate aerobe

(3) Obligate anaerobe

(4) Facultative aerobe

95. State True **(T)** or False **(F)** to the given statements.
Statement (A): All protozoans are heterotrophs and live as predators or parasites.
Statement (B): Viruses did not find a place in the five-kingdom classification of Whittaker.

	(A)	**(B)**
(1)	F	F
(2)	F	T
(3)	T	F
(4)	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

96. Select the **correct** features w.r.t. *Rhizopus*.
(a) Multinucleate and aseptate mycelium is found.
(b) Sexual reproduction takes place by fusion of zoospores.
(c) Formation of female sex organ called ascogonium.
(d) Zygospore is the site of meiosis.
The **correct** ones are

(1) (a), (b) and (c) only

(2) (b) and (d) only

(3) (a) and (d) only

(4) All are correct except (c)

97. Which of the following sequences of mRNA can act as a translational unit and would synthesise a polypeptide having maximum number of amino acids?

(1) 5′-AUGGACACCCAGCGACCCCACACCGC A-3′

(2) 3′-AUGCAGAUUUCAACCCCAAGGGUAAU G-5′

(3) 3′-UGACACCACGCAGCAACCAGCAUG–5′

(4) 5′-AUGCAACCCGACGGGAUUUAGGCAAG G-3′

98. Which of the following processes is ATP or GTP independent?

(1) Aminoacylation of tRNA

(2) Binding of 30S - mRNA complex with $tRNA_{met}$

(3) Binding of mRNA with smaller subunit of ribosomes

(4) Binding of release factor to termination codon

99. RNA polymerase III transcribes the RNA that

(1) Aids in splicing of primary transcript

(2) Is a template molecule of translation

(3) Aids in initiation of translation by recognising the mRNA

(4) Is a precursor of mRNA

100. All of the following structural features contributes to the stability of dsDNA in comparison to RNA, **except**

(1) Presence of thymine in place of uracil

(2) H-bonds

(3) Sugar-phosphate backbone

(4) Stacking of base pairs over each other

101. Number of nucleotides present in the genome of φ × 174 bacteriophage is

(1) 5386

(2) 48502

(3) 4.6×10^6

(4) 3.3×10^9

102. Select the **correctly matched** pair w.r.t organisms employed in commercial production of different products.

(1) *Clostridium butylicum*	–	Acetic acid
(2) *Aspergillus niger*	–	Gluconic acid
(3) *Streptococcus*	–	Amylases
(4) *Acetobacter aceti*	–	Butyric acid

(1) (1)

(2) (2)

(3) (3)

(4) (4)

103. Toddy is obtained from the fermentation of

(1) Sap from palm

(2) Sap of coconut

(3) Fruit juices

(4) Malted cereals

104. Dachigam National Park which is situated in Jammu and Kashmir is home to

(1) Rhinoceros

(2) Snow leopard

(3) Musk deer

(4) Elephants

105. Read the following statements and choose the **correct** option.
Statement A: The Nile perch was introduced in Lake Victoria led to the extinction of indigenous catfishes.
Statement B: Invasive species like *Parthenium* and *Lantana* are threat to indigenous species *Eichhornia*.

(1) Only statement B is incorrect

(2) Only statement A is incorrect

(3) Both statement A and statement B are incorrect

(4) Both statement A and statement B are correct

106. Match column I and column II w.r.t. meninges present in human brain.

	Column I		**Column II**
a.	Dura mater	(i)	Thin, middle layer
b.	Pia mater	(ii)	Outermost layer
c.	Arachnoid	(iii)	Innermost layer

Choose the **correct** option.

(1) a(ii), b(iii), c(i)

(2) a(i), b(ii), c(iii)

(3) a(iii), b(i), c(ii)

(4) a(ii), b(i), c(iii)

107. Brain stem of the human brain consists of

(1) Forebrain, cerebellum and pons

(2) Midbrain, pons and medulla oblongata

(3) Corpus callosum, amygdala and hippocampus

(4) Thalamus, hypothalamus and cerebrum

108. How many of the actions given in the box below involve both locomotion and movement of animals?

Search of food, search of shelter, search of mate, peristalsis, swallowing

Choose the **correct** option.

(1) Four

(2) Three

(3) Two

(4) Five

109. Select the **correct** option to complete the analogy.
Uninucleated muscle fibres : Heart :: Multi-nucleated muscle fibres : ________

(1) Stomach

(2) Intestine

(3) Biceps

(4) Urinary bladder

110. Read the statements given below and select the **correct** option.
Statement I: Water current in the water canal system of sponges is maintained by ciliary movement.
Statement II: The passage of ovum in oviduct is facilitated by ciliary movement.

(1) Both statements I and II are incorrect

(2) Only statement I is correct

(3) Only statement II is correct

(4) Both statements I and II are correct

111. A 15 yr old individual suffered a fracture due to a minor fall. His bone mass was diagnosed to be below the optimum level. Which hormone and vitamin should be given to him during the treatment for increasing his bone mass?

(1) Thyrocalcitonin and vitamin D

(2) Parathormone and vitamin C

(3) Epinephrine and vitamin K

(4) MSH and vitamin A

112. A bony cavity called sella turcica houses ‘X’ and ‘Y’ forms the basal part of diencephalon.
Identify ‘X’ and ‘Y’ respectively and choose the correct option.

(1) Pituitary gland; Hypothalamus

(2) Hypothalamus; Pineal gland

(3) Pineal gland; Pituitary gland

(4) Thyroid gland; Thymus gland

113. A part of renal tubule in which conditional reabsorption of Na^+ and H_2O takes place is

(1) DCT

(2) PCT

(3) Collecting duct

(4) Bowman’s capsule

114. Complete the analogy by selecting the correct option w.r.t. contraction of muscles and change in the volume of thoracic cavity during normal breathing.
Contraction of diaphragm : Antero-posterior axis : : Contraction of external inter-costal muscles : ________

(1) Antero-posterior axis

(2) Dorso-lateral axis

(3) Dorso-ventral axis

(4) Ventro-lateral axis

115. Incomplete cartilaginous rings are absent in all of the given components of conducting part of our respiratory system, **except**

(1) Initial bronchioles

(2) Terminal bronchioles

(3) External nostrils

(4) Pharynx

116. Read the following Assertion **(A)** and Reason **(R)** and choose the **correct** option
Assertion (A): In an aquatic ecosystem pyramid of biomass is inverted.
Reason (R): Biomass of zooplanktons is higher than that of phytoplanktons as life span in former is longer and the latter multiply much faster though having shorter life span.

(1) (A) is true but (R) is false.

(2) Both (A) and (R) are false.

(3) Both (A) and (R) are true and (R) is correct explanation of (A).

(4) Both (A) and (R) are true but (R) is not correct explanation of (A).

117. Grasses → Grasshopper → Birds → Hawk.
What percent of energy is available to birds if at T_4 level energy is 40 J?

(1) 4 kJ

(2) 0.4 kJ

(3) 4 J

(4) 400 kJ

118. Read the following **Assertion (A)** and **Reason (R)** and select the **correct** option.
Assertion (A) : The quasi-fluid nature of lipids, enables lateral movement of proteins within the overall bilayer.
Reason (R): The ability of proteins to move within the membrane is measured as its fluidity.

(1) (A) is false but (R) is true

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

119. Identify the **incorrect** statements w.r.t. microbodies.

(1) These are single membrane-bound minute vesicles that contain various enzymes.

(2) They are associated with oxidation reactions other than those of respiration.

(3) These are found in both plant and animal cells

(4) They possess their own DNA.

120. Which face of Golgi complex gives rise to the secretory vesicles?

(1) *Cis* face

(2) Convex face

(3) Forming face

(4) *Trans* face

121. Number of meiotic divisions required to produce 100 seeds in wheat plant is

(1) 100

(2) 50

(3) 125

(4) 25

122. Read the following statements and select the **correct** option.
Statement A: Intrameiotic interphase is a metabolic stage between telophase I and prophase II.
Statement B: There is no replication of DNA occur during interkinesis.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

123. The most dramatic period of the cell cycle that involves a major reorganisation of virtually all components of the cell is

(1) G_1 phase

(2) M-phase

(3) S-phase

(4) G_0 phase

124. Living pteridophytes are restricted to narrow geographical regions because

(1) Their main plant body is a sporophyte

(2) They possess well-differentiated vascular tissues

(3) Their gametophytes require cool, damp and shady places to grow and need water for fertilization

(4) They bear strobili or cones

125. Identify the unicellular alga from the following.

(1) *Volvox*

(2) *Chlorella*

(3) *Ulothrix*

(4) *Fucus*

126. Which of the given statements about bulliform cells is **incorrect**?

(1) They are found along the veins

(2) These are usually present on abaxial epidermis

(3) They are empty, colourless cells

(4) They can occur in groups

127. Match the column I with column II and select the **correct** option.

	Column I		Column II
a.	Pericycle absent	(i)	Dicot stem
b.	Ring like arrangement of vascular bundles	(ii)	Monocot root
c.	Small and inconspicuous pith	(iii)	Monocot stem
d.	Xylem bundles are said to be polyarch	(iv)	Dicot root

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iii), b(i), c(iv), d(ii)

128. Casparian strips are seen in

(1) Endodermis

(2) Pericycle

(3) Pith

(4) Epidermis

129. Arrange the following substrate in descending order w.r.t. their RQ values
(a) Glucose
(b) Tripalmitin
(c) Proteins
(d) Malic acid
Choose the **correct** option

(1) a > d > c > b

(2) d > c > a > b

(3) c > b > a > d

(4) d > a > c > b

130. Match the **column-I** with **column-II** and choose the **correct** option.

	Column-I		Column-II
(a)	PEPcase	(i)	Embedded in the thylakoid membrane
(b)	ATP synthase	(ii)	Present in mesophyll cell of C_4 plant
(c)	Assimilatory power	(iii)	ATP and $NADPH_2$
(d)	Carotenoids	(iv)	Prevent photo-oxidation of chlorophyll

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(iii), c-(iv), d-(i)

(3) a-(ii), b-(i), c-(iii), d-(iv)

(4) a-(iii), b-(ii), c-(i), d-(iv)

131. How many of the given statements is/are **correct** w.r.t. platelets in humans?
(A) They are nucleated cells produced from megakaryocytes.
(B) Blood contains 1,500,00–3,500,00 platelets mm^{-3}.
(C) Their reduction leads to excessive blood loss from the body on injury.
(D) They are categorised as formed elements along with erythrocytes and leucocytes.
Select the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

132. The middle layer of wall of blood vessels which is comparatively thick in arteries than in veins, is made of

(1) Squamous endothelium

(2) Fibrous connective tissue with collagen fibres

(3) Smooth muscles and elastic fibres

(4) Fibrous connective tissue and squamous endothelium

133. Select the **correct** path of flow of blood in systemic circulation of human heart.

(1) Left ventricle → Pulmonary artery → Body tissues → Pulmonary vein → Right atrium

(2) Right atrium → Right ventricle → Pulmonary artery → Pulmonary vein → Left atrium

(3) Left ventricle → Aorta → Body tissues → Vena cava → Right atrium

(4) Right atrium → Right ventricle → Aorta → Lungs → Vena cava → Left atrium

134. Select the **incorrectly** matched option w.r.t. the mode of excretion (column I), group of animals (column II) and organ/method of excretion (column III).

	Column I	**Column II**	**Column III**
(1)	Ammonotelism	All amphibians	Diffusion across body surface
(2)	Ureotelism	Mammals	Kidneys
(3)	Uricotelism	Reptiles	Kidneys
(4)	Ammonotelism	Many bony fishes	Gills

(1) (1)

(2) (2)

(3) (3)

(4) (4)

135. Motor messages from CNS to urinary bladder result in release of urine. Which of the following events occur during this process?

(1) Simultaneous contraction of both smooth muscles of urinary bladder and urethral sphincter.

(2) Simultaneous relaxation of both skeletal muscles of urinary bladder and urethral sphincter.

(3) Contraction of smooth muscles of urinary bladder and simultaneous relaxation of urethral sphincter.

(4) Contraction of skeletal muscles of urinary bladder and simultaneous relaxation of urethral sphincter.

136. Consider a chemical reaction with optimum temperature of 35°C. If during the reaction, the temperature of the setup is changed from 35°C to 25°C, then the rate of reaction becomes 'X'. Thus, the initial rate of reaction would have been

(1) $\frac{X}{2}$

(2) $\frac{X}{4}$

(3) 2X

(4) 4X

137. The ring structure of glucose can be differentiated from that of 2′ deoxyribose as the former is/has

(1) Homocyclic with 6 carbon atoms in its ring

(2) A homocyclic pentose sugar

(3) Molecular formula of $C_5H_{10}O_5$

(4) Heterocyclic with 5 carbons in its ring

138. Which of the following is true about haem which is the part of the active site of the enzyme, 'peroxidase'?

(1) It is not required for the normal catalytic activity of peroxidase.

(2) It forms a strong and permanent coordination bond with the side chains at the active site of peroxidase and product of the reaction.

(3) Its association with the apoenzyme is transient during catalysis.

(4) It is an organic compound and is tightly bound to the apoenzyme.

139. If guanine makes 35% of the B-DNA molecule, what will be the percentage of substituted pyrimidines present in it?

(1) 70%

(2) 35%

(3) 50%

(4) 15%

140. A student has samples of certain compounds which are listed below:
A – Collagen
B – Glycogen
C – Palmitic acid
D – DNA
E – Calcium ions
His mentor asked him to arrange the category to which these compounds belong to in the increasing order w.r.t. their per cent in the total cellular mass. Which among the following will be the correct sequence?

(1) B < A < C < D < E

(2) E < C < B < D < A

(3) A < B < D < C < E

(4) E < A < C < D < B

141. **Assertion (A) :** Out of all the types of connective tissues, blood is the main circulating fluid that helps in the transport of various substances across the body.
Reason (R) : Blood is a loose connective tissue in which fibroblasts are absent.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) is true but (R) is false

(3) Both (A) and (R) are false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

142. **Assertion (A):** Repeated activation of muscles for a prolonged duration lead to fatigue.
Reason (R): Anaerobic breakdown of glycogen leads to accumulation of lactic acid.
In the light of above statements, select the **correct** option.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are false

143. Choose the descendants of Psilophyton which were abundant during the Jurassic period.

(1) Horsetails

(2) Progymnosperms

(3) Gnetales

(4) Conifers

144. Read the given statements and select the **correct** option.
Statement (A): In an experiment conducted by S. L. Miller in 1953, biomolecules that join to each other *via* peptide bonds during polymerisation were found.
Statement (B): S. L. Miller created electric discharge in a closed flask containing CH_4, NH_3, H_2 and water vapour at 800°F.

(1) Both statements (A) and (B) are incorrect

(2) Only statement (A) is correct

(3) Only statement (B) is correct

(4) Both statements (A) and (B) are correct

145. In a population of 1000 individuals, 25% exhibit a recessive phenotype. What is the frequency of the recessive allele?

(1) 0.05

(2) 0.25

(3) 0.5

(4) 0.95

146. Comprehend the given statements.
(a) Progestasert as well as Multiload-375 prevent conception by inhibiting oogenesis and ovulation.
(b) Contraceptives are regular requirements for the maintenance of reproductive health.
(c) Administration of IUDs within 72 hours of implantation have been found to be effective as an emergency contraceptive.
Select the **correct** option.

(1) Only (a) and (b) are true

(2) Only (a) and (c) are false

(3) (a), (b) and (c) are false

(4) Only (b) and (c) are true

147. **Assertion (A):** Contraceptive injections and diaphragm share the similar mode of action for preventing contraception.
Reason (R): Both inhibit ovulation and implantation as well as alter the quality of cervical mucus to retard the entry of sperms.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

148. Read the following statements w.r.t. humans.
(a) Embryogenesis starts in fallopian tube and embryo enters in uterine cavity at blastocyst stage.
(b) Implantation takes place within the endometrium layer of uterus.
(c) The inner cell mass gets attached to the endometrium and differentiates into the embryo.
Select the option representing **correct** statement only.

(1) (a) only

(2) (b) only

(3) (c) only

(4) (a), (b) and (c)

149. State whether the following statements are **true (T)** or **false (F)** w.r.t. human males.
A. The reproductive system is located in the pelvis region except the external genitalia.
B. The penis is the male external genitalia whose enlarged end is covered by a loose fold of skin called the glans penis.
C. The testes are situated outside the abdominal cavity within a pouch called scrotum.
D. Scrotum helps in maintaining higher temperature than the normal body temperature as it is necessary for fertilization.
Select the **correct** option.

	A	**B**	**C**	**D**
(1)	T	F	T	F
(2)	F	T	F	T
(3)	F	F	T	F
(4)	T	T	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

150. Which of the following is an unpaired structure present in the reproductive system of a human female?

(1) Ovary

(2) Cervix

(3) Mammary gland

(4) Oviduct

151. Read the following statements A and B w.r.t. humans.
Statement A: Penis is made up of special tissue that helps in its erection to facilitate insemination.
Statement B: Seminal plasma is rich in sperms, glucose, calcium and certain enzymes.
Choose the **correct** option.

(1) Both statements A and B are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

152. If a gene encoding the enzyme β-galactosidase is inserted within the selectable marker gene of the plasmid of *E.coli* then, in the presence of a chromogenic substrate

(1) Non-recombinant colonies will produce the blue colour

(2) All transformants will remain white due to inactivation of selectable marker gene

(3) Recombinant colonies will be blue in colour

(4) Non- transformants will appear pink in colour

153. How many of the following is/are the direct method(s) of gene transfer?

Biolistics, Disarmed retrovirus mediated, Micro-injection, Electroporation

Select the **correct** option.

(1) Three

(2) Two

(3) Four

(4) One

154. All of the following are correct for *Eco*RI, **except**

(1) It produces sticky ends after digesting a DNA fragment.

(2) It recognises the recognition sequence, 5' GAATTC 3’.

(3) It is the first restriction enzyme to be isolated from strain Rd of *Escherichia coli*

(4) Its recognition sequence is present in pBR322

155. Which among the following is the part of upstream processing?

(1) Separation and purification of the product

(2) Formulation of preservatives in the product

(3) Strict quality control testing

(4) Maintenance of cells in their physiologically most active phase

156. Oxidation of each molecule of Acetyl-CoA in Krebs cycle produces

(1) One molecule of GTP, three molecules of NADH + H^+ and one $FADH_2$ molecule

(2) One molecule of ATP, three molecules of NADPH + H^+ and one $FADH_2$ molecule

(3) One molecule of ATP, two molecules of NADH + H^+ and three $FADH_2$ molecule

(4) One molecule of ATP, two molecules of NADH + H^+ and two $FADH_2$ molecules

157. How many ATP molecules are directly synthesised in glycolysis from one glucose molecule?

(1) Four

(2) Two

(3) Three

(4) Six

158. Which of the following extrinsic factors is majorly required by plants for the synthesis of protoplasm and act as source of energy?

(1) Light

(2) Temperature

(3) Nutrients

(4) CO_2

159. Select the **incorrectly** matched pair.

(1) Ethylene – Decreases sensitivity of roots to gravity

(2) IAA – Initiate rooting in stem cuttings

(3) GA_3 – Promotes bolting in cabbages

(4) ABA – Promote protein synthesis

160. Presence of berry fruit and axile placentation are characteristic features of plants that belong to the family

(1) Asteraceae

(2) Solanaceae

(3) Poaceae

(4) Malvaceae

161. Single ovule attached to the placenta is found in

(1) Marigold

(2) *Primrose*

(3) Tomato

(4) Pea

162. How many plants in the list given below have zygomorphic flowers?

Canna, Mustard, Datura, Pea, Gulmohur, Bean, *Cassia*, *Petunia*, Ashwagandha

(1) Four

(2) Three

(3) Five

(4) Six

163. Identify the **correct** features of mango fruit.
(a) It develops from monocarpellary superior ovary
(b) It possesses inner stony hard endocarp
(c) Its mesocarp is fibrous and non-edible
(d) Its pericarp is dry and not well differentiated.
Select the **correct** option from below:

(1) (c) and (d) only

(2) (a) and (b) only

(3) (a), (b) and (d) only

(4) All (a), (b), (c) and (d)

164. Read the following statements and select the **correct** option.
Statement A : People affected with haemophilia-B lacks plasma thromboplastin.
Statement B : Cystic fibrosis is an autosomal dominant disorder.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

165. What is the probability of an individual having genotype AaBBCc produced from a cross between AABbCC and AaBbcc?

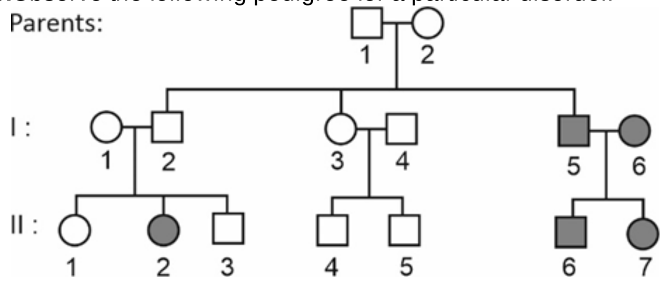
(1) $\frac{1}{16}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{8}$

166. Observe the following pedigree for a particular disorder.



Based on the above pedigree, select the **correct** option.

(1) Parents are homozygous for the trait under study.

(2) The trait under study could be thalassemia.

(3) All the affected individuals are colourblind.

(4) Mating of II 6 with normal female will produce only the affected individuals.

167. Chromosomal complement of person with Turner's syndrome is

(1) 45, X0

(2) 47, XXY

(3) 22, X0

(4) 45, XX

168. In the taxonomic categories which hierarchical arrangement in descending order is **correct** in case of animals?

(1) Kingdom, Order, Class, Phylum, Family, Genus.

(2) Kingdom, Order, Phylum, Class, Family, Genus, Species

(3) Kingdom, Phylum, Class, Order, Family, Genus, Species

(4) Kingdom, Class, Phylum, Family, Order, Genus, Species

169. Read the following statements and choose the **correct** option.
Statement A: Competition can occur between two species, only when resources are limiting.
Statement B: Interspecific competition is more acute than intraspecific competition.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

170. Which of the following methods is used to estimate fishes population in a lake or water body?

(1) Total count

(2) Biomass

(3) Relative density

(4) Indirect count

171. In genetic engineering, elution is a step which involves the

(1) Amplification of desired DNA fragment

(2) Cutting of the desired DNA bands from the agarose gel and their extraction

(3) Separation of DNA fragments over a glass rod in the form of spool

(4) Insertion of rDNA fragments within the host cell

172. Read the following statements and select the **correct** option.

Statement (A): Transgenic mice are being used to test the safety of polio vaccine.

Statement (B): More than 99 % of all the existing transgenic animals are rabbits.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

173. Bioremediation can be defined as the method of

(1) Breeding crops to increase their nutritional value

(2) Regeneration of whole plant from explant

(3) Producing therapeutics for the treatment of diseases

(4) Use of microorganisms to remove pollutants

174. **Assertion (A):** When an antigen enters the body of a person, specific B-cells start dividing to produce antibodies.

Reason (R): If the same pathogen enters our body again, memory B-cells will produce a primary response.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

175. Select the **incorrect** option w.r.t. HIV.

(1) Belongs to group retroviruses

(2) Enveloped virus containing two identical molecules of its genetic material

(3) Does not escape the immune system but predominantly attacks the cells that mediate humoral immune response

(4) There is always a time lag between the infection and appearance of AIDS symptoms

176. Select the **incorrect** statement.

(1) Early Greeks thought that individuals with 'blackbile' belonged to hot personality and would have fevers.

(2) Health could be defined as a state of complete physical, mental and social well-being.

(3) B cells themselves do not secrete antibodies but help T cell to produce them.

(4) The innate defences of our body help to block the entry of pathogens into our body.

177. Select the **correct** match w.r.t structures present in cockroach and their locations.

	Structure	**Location**
(1)	Malpighian tubules	Junction of foregut and midgut
(2)	Gastric caeca	Junction of midgut and hindgut
(3)	Proventriculus	Followed by crop
(4)	Crop	Followed by gizzard

(1) (1)

(2) (2)

(3) (3)

(4) (4)

178. How many of the following animals exhibit blood circulation in which blood is pumped out of the heart and the cells and tissues are directly bathed in it?

Arthropods, Hemichordates, Amphibians, Vertebrates, Reptiles

Select the correct option.

(1) Three

(2) Four

(3) Two

(4) Five

179. Select the option which represents the correct set of differences between bony fishes and cartilaginous fishes.

	Bony fishes	**Cartilaginous fishes**
(1)	Mouth is always located ventrally	Mouth is terminal
(2)	Skin contains placoid scales	Skin is covered with ctenoid scales
(3)	Claspers are absent in males	Pelvic fins of males bear claspers
(4)	Fertilisation is usually internal	Exhibit external fertilisation

(1) (1)

(2) (2)

(3) (3)

(4) (4)

180. Match column I and column II w.r.t structures present in frog.

	Column I		**Column II**
(a)	Forebrain	(i)	Optic lobes
(b)	Hindbrain	(ii)	Cerebellum and medulla oblongata
(c)	Midbrain	(iii)	Sensory papillae
(d)	Cellular aggregations around nerve endings	(iv)	Olfactory lobes

Select the **correct** option.

(1) a-(i), b-(ii), c-(iv), d-(iii)

(2) a-(iii), b-(i), c-(ii), d-(iv)

(3) a-(iv), b-(ii), c-(i), d-(iii)

(4) a-(i), b-(ii), c-(iii), d-(iv)